

Luke Cullen

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science - Minor in Data Science - 3.72 GPA

Orlando, FL

August 2023 – May 2027

EXPERIENCE

Lead Developer

May 2025 – Present

DenTraceX

Orlando, FL

- Non-profit organization dedicated to collecting dental records of missing persons, in order to cross-reference them with unidentified remains
- Authoring a secure database from the ground up to make submitting and accessing data for medical examiners as accessible as possible
- Working with a team of diverse disciplines to process and handle sensitive medical information

Hackathon Organzier

July 2025 – Present

KnightHacks

Orlando, FL

- Working with our directing team to utilize over **\$60,000** worth of event funding.
- Constructing an environment that provides hundreds of student hackers deadline-limited programming experience.
- Assisting in logistics for a target of **1000+** people attending the event.
- Planning engaging events to combat development burnout and make KnightHacks truly memorable.

Cashier

February 2023 – Present

Publix Super Markets

Orlando, FL

- Processing customer transactions efficiently while providing great service.
- Able to adapt easily to changes in working condition/customer inflow.
- Working as a team to ensure the completion of all daily tasks while keeping store operations running smoothly.

PROJECTS

DenTraceX Repo | *React, ExpressJS, TypeScript, PostgreSQL, Docker, Coolify, Minio S3*

May 2025 – Present

- Full-stack implementation of the DenTraceX NGO for processing dental records and exposing them to examiners
- Self-hosted and implemented a **Minio S3** instance to allow for secure access to sensitive client records
- Constructed novel code-based representation for jaw/tooth records, allowing easy searching/filtering to identify **real-world** human remains
- Created an interface to onboard moderators, verify and approve the identities of submissions/examiners, and enforce user permissions

Voiceboard | *React, TypeScript, Gemini API, Mermaid*

September 2024

- Created for ShellHacks 2024, won **1st place** for **Google's Hack for Social Good Challenge**
- Aims to make whiteboarding accessible to all using voice control and Google's **Gemini AI**
- Utilizes Chrome's speech recognition to receive prompts from users
- Implements prompt engineering through the **Gemini API** to create Mermaid syntax
- Utilizes context awareness to allow users to iterate on diagrams

Lootcode | *React, Clerk, TypeScript, MySQL, Zx Shell, Docker, tRPC*

February 2024 – May 2024

- Created for KnightHacks Project Launch 2024, won **1st place**
- Developed a gamified application to study Data Structures and Algorithms
- Uses **Docker containers** to self-host code grading instances, maintaining security of the server
- Implements a **MySQL** database to track user statistics, progress, and game inventory
- **7000+** unique visits, featured in **Linux Magazine's** OSS section

TECHNICAL SKILLS

Languages: Java, Python, C/C++/C#, MySQL, JavaScript, TypeScript, HTML/CSS

Libraries/Frameworks: React, ExpressJS, Next, Prisma, Clerk, tRPC, libGDX, Unity, Node.js, Tailwind CSS, Websockets, OpenCV, MediaPipe, pandas

Developer Tools: Linux, Coolify, Git, Docker, VS Code, Eclipse